

TEST OF HYPOTHESIS IN A MULTIVARIATE HYPERGEOMETRIC
CASE USING A BAYESIAN APPROACH

D.J. de Waal: Department of Mathematical Statistics, University of the Orange Free State, P.O. Box 339, Bloemfontein, 9300 Republic of South Africa.

Key words: Bayes estimation and test of hypothesis; multivariate hypergeometric and inverse multivariate hypergeometric distributions.

Summary: A test of hypothesis situation in a multivariate hypergeometric case is considered from the Bayesian point of view. Under the assumption of a uniform prior on the parameters, the posterior distribution turns out to be an inverse multivariate hypergeometric distribution. Some correspondence between the classical test and the Bayes test has been drawn.

1. INTRODUCTION

Although the theory in this paper will be illustrated by a numerical example one can find many examples of application in the real world. Suppose for instance, that in a certain university with a known number of students N (not large) we wish to estimate the numbers of students $\theta_1, \theta_2, \dots, \theta_r$ in favour of each of r fixed different political parties. A random sample of n students is drawn and the numbers in favour of the different political parties are observed, say $x_1, x_2, \dots, x_r$; $\sum x_i = n$.

Apart from estimating θ_i , $i=1, \dots, r$ through a Bayesian approach we will be looking at a Bayes test for testing the hypothesis that θ_i , $i=1, \dots, r$ are equal to certain pre-assigned values. For illustration purposes, however, we will consider an urn containing a fixed number N of balls of different colours, say r of them, but of unknown numbers $\theta_1, \theta_2, \dots, \theta_r$ respectively. A sample of n balls is drawn without replacement and the numbers of the different colours in the sample are observed. Let $X_1, X_2, \dots, X_r$ denote random variables where X_i is the number of observed balls of the i -th colour such that $\sum_{i=1}^r X_i = n \leq N$.

Then the joint probability distribution of $X_1, \dots, X_r$ is given by

$$(1.1) \quad f(x_1, \dots, x_r | \theta_1, \dots, \theta_r) = \frac{\binom{\theta_1}{x_1} \binom{\theta_2}{x_2} \dots \binom{\theta_r}{x_r}}{\binom{N}{n}}.$$

This is the well-known multivariate hypergeometric distribution. See for instance Patil, *et al.* (1984) pp. 114-115 for some of its properties. We are interested in this paper in the test of the hypothesis

$$H_0: \theta_1 = \theta_2 = \dots = \theta_r$$

against

$$H_1: \text{Not } H_0$$

from a Bayesian point of view.

Before we consider that, however, we will first look at the posterior joint probability distribution of $\theta_1, \dots, \theta_r$.

2. POSTERIOR DISTRIBUTION

Consider the parameters $\theta_1, \dots, \theta_r$ ($\sum \theta_i = N$) as random variables with a uniform prior probability distribution on the parameter

space

$$(2.1) \quad \Theta = \{(\theta_1, \theta_2, \dots, \theta_r): 0 \leq \theta_i \leq N, i=1, 2, \dots, r; \sum \theta_i = N\}$$

given by

$$(2.2) \quad \pi(\theta_1, \dots, \theta_r) = (r-1)! [(N+r-1)(N+r-2)\dots(N+1)]^{-1} \\ = \frac{[N+r-1]^{-1}}{r-1}.$$

The joint posterior probability distribution of $\theta_1, \theta_2, \dots, \theta_r$ given the observed sample $x_1, x_2, \dots, x_r$ is then given by

$$(2.3) \quad \pi(\theta_1, \dots, \theta_r | x_1, \dots, x_r) = \frac{[\theta_1]}{[x_1]} \frac{[\theta_2]}{[x_2]} \dots \frac{[\theta_r]}{[x_r]} \frac{[N+r-1]}{[n+r-1]}$$

where $\theta_i = x_i, x_i+1, \dots, N-n+x_i$; $\sum x_i = n$; $\sum \theta_i = N$.

The distribution (2.3) seems to be unknown in the literature and will be referred to as the inverse multivariate hypergeometric probability distribution. It differs from the multivariate inverse hypergeometric distribution given in Patil, et al. (1985) pp. 115-116. The constant term $1/[n+r-1]$ in (2.3) is derived by induction knowing that for $r=2$ the term is $1/[n+1]$ (see Patil, et al. (1985) p. 48). The moments of (2.3) can be obtained by using the following results which are easy to obtain:

$$(2.4) \quad E(\theta_i | x_1, \dots, x_r) = \frac{(x_i+1)(N+r)}{n+r} - 1$$

$$(2.5) \quad E((\theta_i+1)(\theta_j+1) | x_1, \dots, x_r) = \frac{(x_i+1)(x_j+1) \frac{[N+r+1]}{[n+r+1]}}{\frac{[N+r-1]}{[n+r-1]}}$$

$$(2.6) \quad E((\theta_i+1)(\theta_i+2) | x_1, \dots, x_r) = \frac{(x_i+1)(x_i+2) \frac{[N+r+1]}{[n+r+1]}}{\frac{[N+r-1]}{[n+r-1]}}$$

Consider the following example:

Example 1

Let $N=15$ and $n=7$. Suppose $r=3$ and $x_1=4$, $x_2=2$ and $x_3=1$ have been observed. Then the joint posterior probability distribution of θ_1 and θ_2 given $x_1=4$, $x_2=2$, $x_3=1$ is given by

$$\pi(\theta_1, \theta_2, \theta_3 | 4; 2; 1) = \frac{\begin{bmatrix} \theta_1 \\ 4 \end{bmatrix} \begin{bmatrix} \theta_2 \\ 2 \end{bmatrix} \begin{bmatrix} \theta_3 \\ 1 \end{bmatrix}}{\begin{bmatrix} 17 \\ 9 \end{bmatrix}}$$

$$\theta_1 = 4, 5, \dots, 12; \theta_2 = 2, 3, \dots, 10; \theta_3 = 1, 2, \dots, 9; \theta_1 + \theta_2 + \theta_3 = 15.$$

See table 1, column (i) in the appendix for the tabulated probability distribution.

The moments are:

$$E(\theta_1 | 4; 2; 1) = 8, \quad \text{Var}(\theta_1 | 4; 2; 1) = 3,2727,$$

$$E(\theta_2 | 4; 2; 1) = 4,4, \quad \text{Var}(\theta_2 | 4; 2; 1) = 2,74909,$$

$$\text{Cov}(\theta_1, \theta_2 | 4; 2; 1) = -1,9636.$$

A number of conclusions can be drawn from the posterior probabilities and the moments as for instance:

1. A uniform prior distribution on θ is assumed which means that we have vague information about the parameters. That

means we put a probability $\frac{\begin{bmatrix} 17 \\ 2 \end{bmatrix}^{-1}}{\begin{bmatrix} 17 \\ 9 \end{bmatrix}} = 0,00735$ on each point in θ . A multinomial distribution on θ will put much more specific prior information on the parameters.

2. The Bayes estimates of θ_1 and θ_2 under squared error loss are given by

$$E(\theta_1 | 4; 2; 1) = 8$$

and

$$E(\theta_2 | 4; 2; 1) = 4,4$$

respectively. The obvious estimates are, however, at the mode of the posterior distribution, i.e. where $\theta_1 = 9$ and $\theta_2 = 4$ (see table 1(i)).

3. The posterior probability that the three types of balls in the urn are evenly distributed, i.e. $\theta_1 = 5$, $\theta_2 = 5$, $\theta_3 = 5$, is given by

$$\pi(5;5;5|4;2;1) = 0,01028.$$

If we compare this with the largest probability 0,06220 which occurs at $\theta_1 = 9$ and $\theta_2 = 4$, it is more than 6 times smaller, making it therefore quite unlikely that the balls are evenly distributed. The probability that most balls are of the first type is

$$P(\theta_1 > \max(\theta_2, \theta_3) | 4;2;1) = 0,7951$$

which means that we are almost 80% sure that the first type of balls are more than the second and more than the third.

4. If we would like to test the hypothesis that the balls are evenly distributed then, according to remarks 2 and 3, it seems quite unlikely that we will accept the hypothesis. The posterior probability $\pi(5;5;5|4;2;1)$ has arisen however from the prior probability $P(\theta_1 = 5, \theta_2 = 5) = 0,00735$ to 0,01028 which means that the information from the sample supports the hypothesis that the balls are evenly distributed. We may, however, have quite strong evidence *a priori* that $\theta_1 = \theta_2 = \theta_3$, which may be the reason for looking into the hypothesis of equally distributed colours. Say for instance that these urns are filled by a machine which is supposed to fill them with 5 red balls (type 1), 5 white balls (type 2) and 5 green balls (type 3). The urn can only take $N = 15$ balls so that there can be no

question about the number of balls in the urn, but the machine does sometimes make a mistake in the numbers of the colours. In this example we have strong prior knowledge in the event $\{\theta_1 = 5, \theta_2 = 5, \theta_3 = 5\}$ and it would be unreasonable to put a uniform distribution on θ . This brings us to the case of the test of hypothesis with some knowledge about the truth of the hypothesis.

5. From table 1(i) we find that the set $\theta_0 \subset \theta$ where the posterior probabilities have become smaller than the prior probability of 0,00735 is given by

$$\theta_0 = \{(\theta_1; \theta_2; \theta_3) : (\theta_1; \theta_2) = (4;3), (4;4), (4;5), (4;6), (4;7), (4;8), (4;9), (4;10), (5;2), (5;3), (6;2), \theta_3 = 15 - \theta_1 - \theta_2\}.$$

The set θ_0^c is called the Bayes confidence set for the parameters under the uniform prior $\pi(\theta_1, \theta_2, \theta_3) = \begin{bmatrix} 17 \\ 2 \end{bmatrix}^{-1}$ and given the observation $x_1 = 4, x_2 = 2, x_3 = 1$.

3. TEST OF HYPOTHESIS

Let the hypothesis we are interested in be

$$H_0: \theta_1 = \theta_2 \dots \theta_r = \theta_0 \text{ (say)}$$

against

$$H_1: \text{Not } H_0.$$

Suppose we have prior knowledge about the occurrence of H_0 and place a mass $p(0 < p \leq 1)$ on the parameter space

$$\omega = \{(\theta_1, \dots, \theta_r) : \theta_1 = \theta_2 = \dots = \theta_r\}$$

which is a subspace of θ . Denote the probability by

$$\pi_0(\omega) = p.$$

The rest of the mass $(1-p)$ is spread uniformly over the space

ω^C . Denote the resulting distribution by

$$\pi_1(\theta_1, \theta_2, \dots, \theta_r) = (1-p) \left[\binom{N+r-1}{r-1} - 1 \right]^{-1}, (\theta_1, \theta_2, \dots, \theta_r) \in \omega^C.$$

The joint posterior probability distribution of $\theta_1, \dots, \theta_r$ on ω and ω^C are known (see de Groot (1970) p. 238):

Posterior probability on ω :

$$\begin{aligned} (3.1) \quad \pi_0(\theta_1 = \theta_2 = \dots = \theta_r | x_1, x_2, \dots, x_r) \\ &= \frac{p f(x_1, \dots, x_r | \omega)}{p f(x_1, \dots, x_r | \omega) + (1-p) \sum_{\omega^C} f(x_1, \dots, x_r | \theta_1, \dots, \theta_r) \pi_1(\theta_1, \dots, \theta_r)} \\ &= \frac{p \begin{bmatrix} \theta_0 \\ x_1 \end{bmatrix} \begin{bmatrix} \theta_0 \\ x_2 \end{bmatrix} \dots \begin{bmatrix} \theta_0 \\ x_r \end{bmatrix} / \begin{bmatrix} N \\ n \end{bmatrix}}{p \begin{bmatrix} \theta_0 \\ x_1 \end{bmatrix} \begin{bmatrix} \theta_0 \\ x_2 \end{bmatrix} \dots \begin{bmatrix} \theta_0 \\ x_r \end{bmatrix} / \begin{bmatrix} N \\ n \end{bmatrix} + (1-p) \sum_{\omega^C} \begin{bmatrix} \theta_1 \\ x_1 \end{bmatrix} \begin{bmatrix} \theta_2 \\ x_2 \end{bmatrix} \dots \begin{bmatrix} \theta_r \\ x_r \end{bmatrix} / \begin{bmatrix} N \\ n \end{bmatrix} \left\{ \binom{N+r-1}{r-1} - 1 \right\}} \\ &= p \begin{bmatrix} \theta_0 \\ x_1 \end{bmatrix} \begin{bmatrix} \theta_0 \\ x_2 \end{bmatrix} \dots \begin{bmatrix} \theta_0 \\ x_r \end{bmatrix} / (Q_1 + Q_2) \end{aligned}$$

where

$$\begin{aligned} Q_1 &= p \begin{bmatrix} \theta_0 \\ x_1 \end{bmatrix} \begin{bmatrix} \theta_0 \\ x_2 \end{bmatrix} \dots \begin{bmatrix} \theta_0 \\ x_r \end{bmatrix} \\ Q_2 &= (1-p) \left\{ \binom{N+r-1}{r-1} - \begin{bmatrix} \theta_1 \\ x_1 \end{bmatrix} \begin{bmatrix} \theta_2 \\ x_2 \end{bmatrix} \dots \begin{bmatrix} \theta_r \\ x_r \end{bmatrix} \right\} / \left\{ \binom{N+r-1}{r-1} - 1 \right\}. \end{aligned}$$

Posterior probability distribution on ω^C

$$\begin{aligned} (3.2) \quad \pi_1(\theta_1, \dots, \theta_r | x_1, x_2, \dots, x_r) \\ &= (1-p) \begin{bmatrix} \theta_1 \\ x_1 \end{bmatrix} \begin{bmatrix} \theta_2 \\ x_2 \end{bmatrix} \dots \begin{bmatrix} \theta_r \\ x_r \end{bmatrix} / \{Q_1 + Q_2\} \left\{ \binom{N+r-1}{r-1} - 1 \right\} \\ & \quad (\theta_1, \dots, \theta_r) \in \omega^C. \end{aligned}$$

A lot of information can now be obtained from (3.1) and (3.2). We could, of course, use another distribution rather than the uniform for the spreading of the mass of $1-p$ over ω^C if we had more information.

We will, however, use the uniform for the example considered in section 2.

Example 2

Consider example 1 again. Let

$$H_0: \theta_1 = \theta_2 = \theta_3 = 5$$

$$H_1: \text{Not } H_0.$$

Assume that we know that it is quite likely for H_0 to be true. We know for instance that the machine which fills the urns is supposed to fill them with 5 red, 5 white and 5 green balls. It might for example be reasonable to let $p = \frac{1}{2}$ in (3.1) and (3.2). We are therefore not giving the null hypothesis more credit to be true than the alternative. We therefore start off with the prior probability distributions of

$$\pi_0(\theta_1 = \theta_2 = \theta_3) = \frac{1}{2}, \quad (\theta_1, \theta_2, \theta_3) \in \omega$$

$$\pi_1(\theta_1, \theta_2, \theta_3) = \frac{1}{2} \left[\begin{bmatrix} 17 \\ 2 \end{bmatrix} - 1 \right]^{-1}, \quad (\theta_1, \theta_2, \theta_3) \in \omega^C.$$

Then the posterior probability distribution according to (3.1) and (3.2) becomes

$$\begin{aligned} \pi_0(5;5;5 | 4;2;1) &= \frac{\begin{bmatrix} 5 \\ 4 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix}}{\begin{bmatrix} 5 \\ 4 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix} + \left[\begin{bmatrix} 17 \\ 9 \end{bmatrix} - \begin{bmatrix} 5 \\ 4 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix} \right] / \left[\begin{bmatrix} 17 \\ 2 \end{bmatrix} - 1 \right]} \\ &= 0,58381 \end{aligned}$$

$$\pi_1(\theta_1, \theta_2, \theta_3 | 4; 2; 1) = \frac{\frac{1}{2} \begin{bmatrix} \theta_1 \\ 4 \end{bmatrix} \begin{bmatrix} \theta_2 \\ 2 \end{bmatrix} \begin{bmatrix} \theta_3 \\ 1 \end{bmatrix} / \left[\begin{bmatrix} 17 \\ 2 \end{bmatrix} - 1 \right]}{428,2222}, (\theta_1, \theta_2, \theta_3) \in \omega^C.$$

The probability distribution is tabulated in table 1(ii) in the appendix.

Remarks

1. Note that the posterior probability for $\theta_1 = \theta_2 = \theta_3 = 5$ has risen to 0,58381 from the prior of $p = 0,5$. In fact, any value of p will result in a larger posterior probability for this event than the prior probability. This means that the data $x_1 = 4, x_2 = 2, x_3 = 1$ will support any prior knowledge. One may look at the subset of the sample space S that will support the prior probability on the subset ω .

This subset is found to be

$$S_0 = \{(x_1, x_2, x_3) : (x_1, x_2) = (1; 2), (1; 3), (1; 4), (2; 1), (2; 2), (2; 3), (2; 4), (3; 1), (3; 2), (3; 3), (4; 1), (4; 2), x_1 + x_2 + x_3 = 7\}.$$

The critical Bayes set is S_0^C where $S = \{(x_1, x_2, x_3) : 0 \leq x_1, x_2, x_3 \leq 7; x_1 + x_2 + x_3 = 7\}$.

2. The P -value for the observation $x_1 = 4, x_2 = 2, x_3 = 1$, i.e. the sum of the probabilities under H_0 of obtaining this event (0,03885) and events with probabilities equal to or smaller than 0,03885. This gives $P = 0,30069$ which means we cannot reject H_0 at a 5% (not even a 30%) level. Denote the subset of the sample space that contributes to the P -value by S_p , then the subset of points that do not contribute to the P -value is $S_p^C = \{(x_1, x_2, x_3) : (x_1, x_2) = (1; 3), (2; 2), (2; 3), (3; 1), (3; 2), (3; 3), x_1 + x_2 + x_3 = 7\}$.

The acceptance region S_A using the classical statistical test procedure is the following:

$$S_A = S_0 \cup \{(1;1;5), (5;1;1), (1;5;1)\}.$$

Criterion:

If at least one ball of each colour is observed, then we accept the hypothesis that the balls are evenly distributed. The probability of the type I error is given by 0,0559 which conforms closely to the usual 5% value.

3. The posterior probability that the red balls are more than the white or the green balls is 0,3344 while the posterior probability that white is the most, is 0,04299. We notice that there is a $0,58381 + 0,3344 = 0,91821$ probability that the red balls are just as many or more than the other colours in the container. Hence although the P -value might lead us to assume that the balls are evenly distributed, the posterior distribution indicates that there is only an 8% chance that white or green outnumber red.
4. We can obtain a Bayes confidence set on $(\theta_1, \theta_2, \theta_3)$ under the prior information stated in H_0 and H_1 from table 1(ii). The set $e_I \subset e$ where the posterior probabilities decrease from the prior probabilities 0,5 and 0,003704 under H_0 and H_1 respectively follows from table 1(ii) as

$$e_I = e_0 \cup \{(5;4;6), (5;9;1), (7;2;6)\},$$

where e_0 is given in remark 5 of section 2. Thus e_I^C provides a smaller confidence region for the parameters than e_0^C . The reason is that we have more prior information on

$(\theta_1, \theta_2, \theta_3)$ than in the previous case.

5. We described in this section Bayesian inferences on the two hypotheses H_0 and H_1 . This is in fact what a Bayesian is doing instead of testing a hypothesis as in the classical test of statistics inferences, but much more information is available from the Bayesian analysis.

4. SPECIAL CASES

Noel Cressie & Allan Seheult (1985) consider the estimation of θ in the hypergeometric case, namely

$$f(x|\theta) = \frac{\binom{\theta}{x} \binom{N-\theta}{n-x}}{\binom{N}{n}}, \quad x = 0, 1, \dots, n$$

which is a special case of the multivariate hypergeometric case considered above. Cressie and Seheult assume a binomial prior distribution on θ . If we assume a uniform prior instead on θ , i.e.

$$\pi(\theta) = \frac{1}{N+1}, \quad \theta = 0, 1, \dots, N$$

then the posterior probability distribution for θ , given $X=x$ has been observed, becomes the inverse hypergeometric probability distribution (see Patil, et al. (1984), p. 48)

$$\pi(\theta|x) = \frac{\binom{\theta}{x} \binom{N-\theta}{n-x}}{\binom{N+1}{n+1}}, \quad \theta = x, x+1, \dots, N-n+x.$$

ACKNOWLEDGEMENT

The assistance of Miss N. de la Rey for the numerical calculations is greatly appreciated. Thanks to the editor and the referees for their careful reading of the paper and valuable comments.

APPENDIX

Table 1: Posterior distributions of $\theta_1, \theta_2, \theta_3$ given $x_1 = 4, x_2 = 2, x_3 = 1$ under(i) a uniform prior $\pi(\theta_1, \theta_2, \theta_3) = 0,00735$ and(ii) a prior under $H_0: \pi_0(5;5;5) = \frac{1}{2}$ and under

$$H_1: \pi_1(\theta_1, \theta_2, \theta_3) = \frac{1}{2}(0,00741)$$

θ		(i)	(ii)	
θ_1	θ_2	$\pi(\theta_1, \theta_2 4; 2; 1)$	$\pi_0(\theta_1 = \theta_2 = 5 4; 2; 1)$	$\pi_1(\theta_1, \theta_2 4; 2; 1)$
4	2	0,00037		0,00016
4	3	0,00099		0,00042
4	4	0,00173		0,00073
4	5	0,00247		0,00104
4	6	0,00309		0,00130
4	7	0,00346		0,00145
4	8	0,00346		0,00145
4	9	0,00296		0,00125
4	10	0,00185		0,00078
5	2	0,00165		0,00069
5	3	0,00432		0,00182
5	4	0,00740		0,00311
5	5	0,01028	0,58381	
5	6	0,01234		0,00519
5	7	0,01296		0,00545
5	8	0,01152		0,00484
5	9	0,00740		0,00311
6	2	0,00432		0,00182
6	3	0,01111		0,00467
6	4	0,01851		0,00778
6	5	0,02468		0,01038
6	6	0,02777		0,01168
6	7	0,02592		0,01090
6	8	0,01728		0,00727
7	2	0,00864		0,00363
7	3	0,02160		0,00908
7	4	0,03455		0,01453
7	5	0,04319		0,01816
7	6	0,04319		0,01816

7	7	0,03023	0,01271
8	2	0,01440	0,00605
8	3	0,03455	0,01453
8	4	0,05183	0,02180
8	5	0,05759	0,02422
8	6	0,04319	0,01816
9	2	0,02073	0,00872
9	3	0,04665	0,01962
9	4	0,06220	0,02615
9	5	0,05183	0,02180
10	2	0,02592	0,01090
10	3	0,05183	0,02180
10	4	0,05183	0,02180
11	2	0,02715	0,01142
11	3	0,04072	0,01713
12	2	0,02036	0,00856

REFERENCES

- CRESSIE, N. & SEHEULT, A. (1985). Empirical Bayes estimation in sampling inspection. *Biometrika* 72, 2, pp. 451-458.
- DE GROOT, M.H. (1970). *Optimal statistical decisions*. McGraw-Hill. N.Y.
- PATIL, G.P., BOSWELL, M.T., JOSHI, S.N. & RATNAPARKHI, M.V. (1984). *Dictionary and classified bibliography of statistical distributions in scientific work*. International Co-operative Publishing House, Fairland, Maryland, U.S.A.

Financially supported by the CSIR and the University of the O.F.S. Central Research Fund.